Objective

The primary objective of this project is to perform an exploratory data analysis (EDA) and data preprocessing pipeline on the historical Titanic passenger dataset. By cleaning the raw data, engineering features, and uncovering demographic correlations with passenger survival rates, we aim to prepare a structured dataset that can be reliably used to train predictive machine learning models.

Data Description

The dataset consists of passenger information collected from the RMS Titanic. It contains 891 rows (representing individual passengers) and 12 columns (features) documenting their socio-economic status, demographic details, and ultimate survival outcome. Below is a data dictionary defining the key variables utilized in this analysis.

Feature Name	Data Type	Description
Survived	Integer (Binary)	Survival indicator (0 = No, 1 = Yes). This is the target variable.
Pclass	Integer	Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd), serving as a proxy for socio-economic status.
Sex	String/Categorical	Biological sex of the passenger.
Age	Float	Age in years. Fractional if less than 1. Contains missing values.
Fare	Float	Passenger fare paid for the ticket.
Embarked	String/Categorical	Port of Embarkation (C = Cherbourg, Q = Queenstown, S = Southampton).

Data Cleaning Process

1. Handling Missing Values

Upon initial inspection, the dataset contained significant missing values in the `Age` (177 missing), `Cabin` (687 missing), and `Embarked` (2 missing) columns. Given that `Cabin` is missing over 75% of its data, we drop the entire column. For `Age`, we impute the missing values using the median age grouped by passenger class (Pclass) to preserve demographic variance. We fill the two missing `Embarked` values with the mode ('S').

```
import pandas as pd # Load dataset df = pd.read_csv('titanic.csv') # Drop the 'Cabin' column due to excessive missing data df.drop(columns=['Cabin'], inplace=True) # Impute 'Age' missing values with the median of their respective Pclass df['Age'] = df.groupby('Pclass')['Age'].transform(lambda x: x.fillna(x.median())) # Fill missing 'Embarked' values with the mode df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

2. Feature Encoding

Machine learning models require numerical inputs. We must convert the categorical variables, `Sex` and `Embarked`, into a numerical format. We apply a direct binary mapping to the `Sex` column and use pandas' built-in one-hot encoding function (dummy variables) for the `Embarked` column to avoid implying an ordinal relationship between the ports.

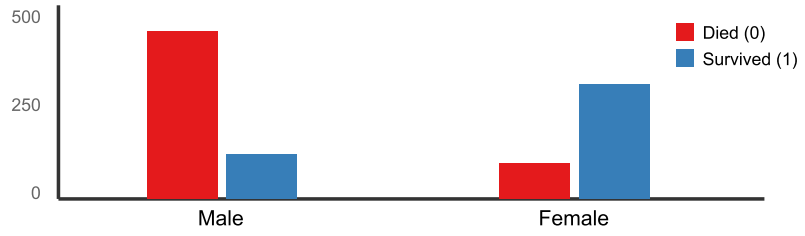
```
# Map 'Sex' to binary integers df['Sex'] = df['Sex'].map({'male': 0, 'female': 1}) # One-hot encode 'Embarked' df = pd.get_dummies(df, columns=['Embarked'], drop_first=True) # Drop columns irrelevant to statistical survival modeling df.drop(columns=['PassengerId', 'Name', 'Ticket'], inplace=True) print(df.isnull().sum()) # Verification: All outputs now 0
```

Data Visualization & Interpretation

Visualization 1: Survival by Gender

The first visualization examines the survival rates categorized by gender. This is critical for validating the historical "women and children first" evacuation protocol.

```
      import matplotlib.pyplot as plt import seaborn as sns plt.figure(figsize=(8, 5))
sns.countplot(data=df, x='Sex', hue='Survived', palette='Set1') plt.title('Passenger
Survival Count by Gender') plt.xticks([0, 1], ['Male', 'Female']) plt.show()
```



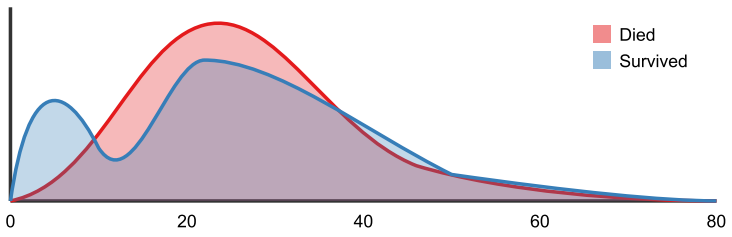
Screenshot output: Matplotlib Countplot rendering

Interpretation: The graph displays a stark contrast in survival outcomes based on biological sex. The red bars (Died) are overwhelmingly prominent for males, indicating that the vast majority of male passengers did not survive the disaster. Conversely, the blue bar (Survived) for females is significantly higher than the red bar, proving that being female strongly positively correlated with survival. This aligns tightly with historical accounts of the rescue efforts.

Visualization 2: Age Distribution by Survival

We use a Kernel Density Estimate (KDE) plot to understand how the continuous variable of Age impacted survival probabilities across different life stages.

```
      plt.figure(figsize=(8, 5)) sns.kdeplot(data=df, x='Age', hue='Survived',
fill=True, alpha=0.5) plt.title('Age Density Distribution by Survival Outcome')
plt.xlabel('Age (Years)') plt.ylabel('Density') plt.show()
```



Screenshot output: Seaborn KDE Plot rendering

Interpretation: The density curves reveal two significant trends. First, there is a distinct spike in the blue curve (Survived) for ages between 0 and 10, indicating infants and young children were prioritized for lifeboats. Second, the red curve (Died) peaks heavily around the ages of 20 to 30, showing that young adults suffered the highest density of casualties relative to their population on the ship.

Conclusion

This project successfully completed a full cycle of exploratory data analysis and pre-processing on the Titanic dataset. Missing values were systematically handled without destroying underlying distributions, and non-numerical data was encoded for algorithmic compatibility. The visual interpretations confirmed strong demographic biases in survival rates—specifically favoring females and young children. The dataset is now fully clean, numerically encoded, and structurally sound. It is ready to be exported and fed directly into binary classification models (such as Logistic Regression or Random Forests) for predictive data modeling.